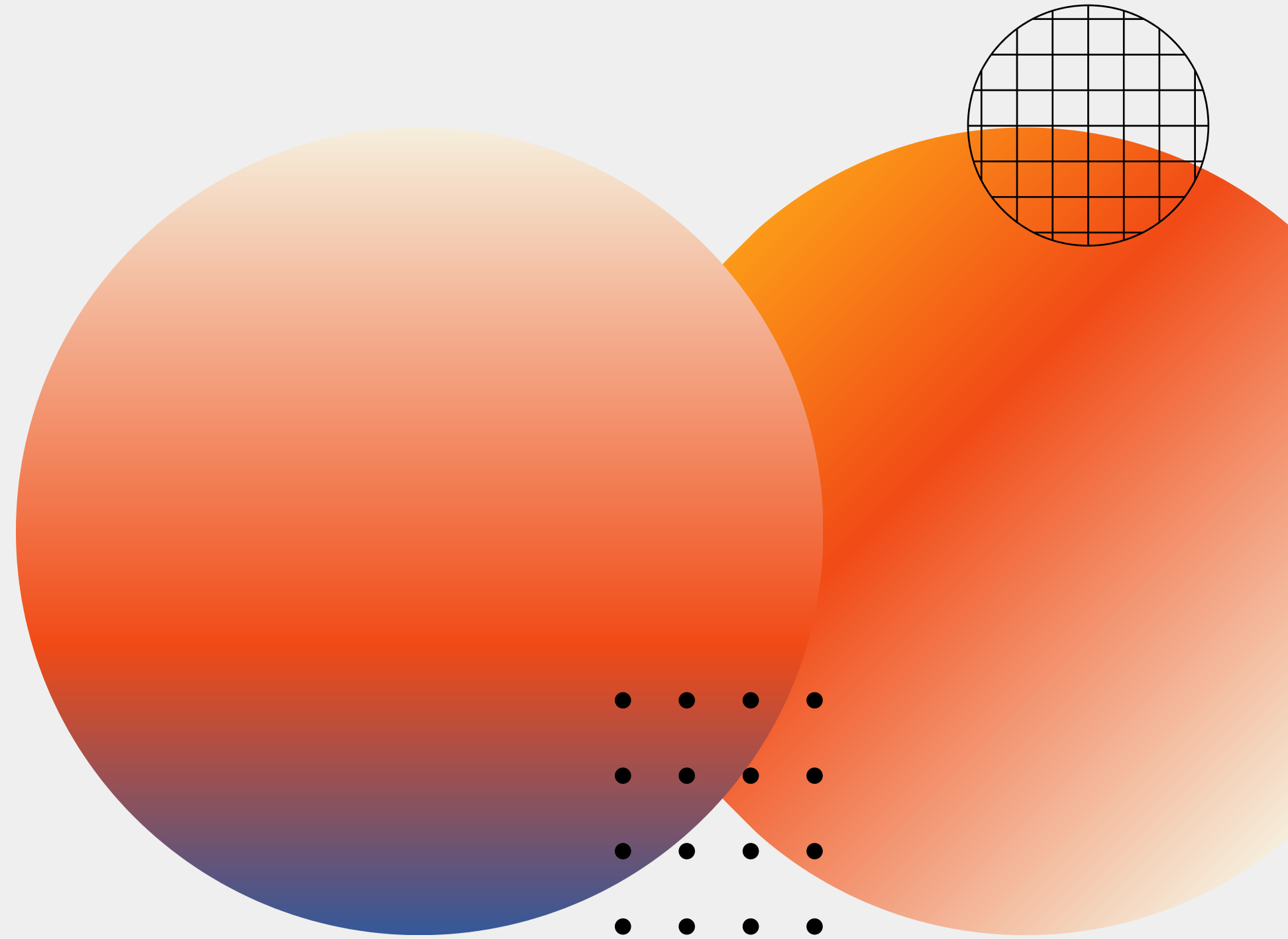


Bivariate Relationships: Contingency Tables



Professor Burrell Vann Jr



Bivariate Relationships

Examining the relationship (association) between two variables

- How changes in the attributes of one variable **affect** changes in the attributes of the other variable
- If two variables are associated, we can use a score for one case on one variable to “predict” the score of that case on the other variable

Two variables known as **dependent** and **independent** variables

Bivariate relationships show us the **strength** of the relationship, as well as the **direction** (or pattern) of the relationship

Statistical significance tests of our bivariate relationships allow us to investigate whether or not the relationships observed in our sample were the result of random **chance**

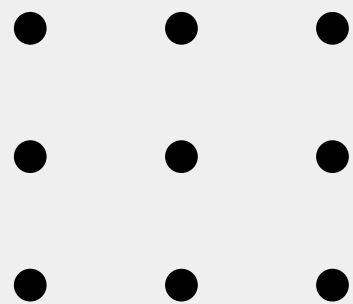
Independent vs. Dependent Variables

Independent

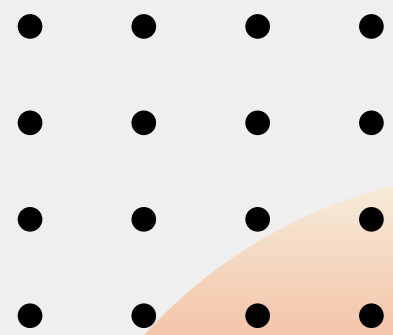
-X Variable
-Predictor

Dependent

-Y Variable
-Outcome

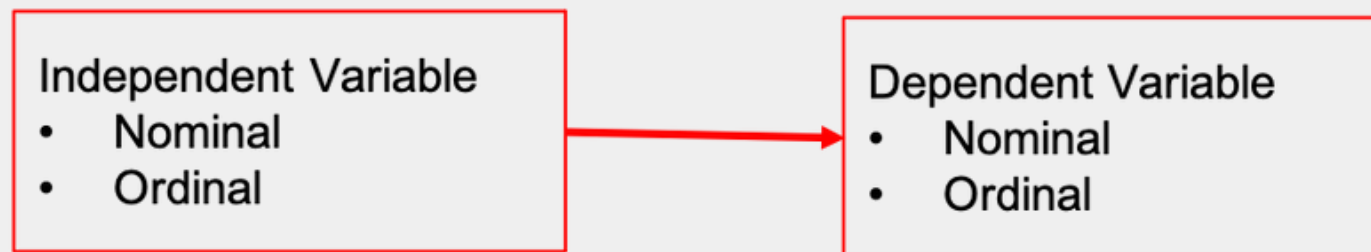


Typical Level of Measurement for Bivariate Pairs

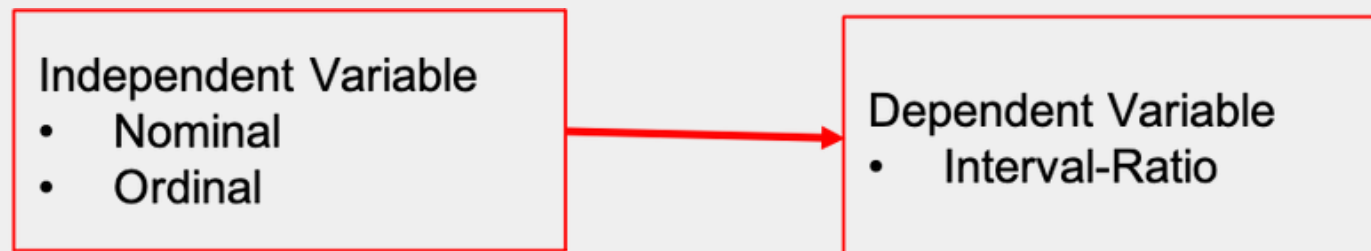


- 1** Independent **categorical** (*nominal or ordinal*) & Dependent **categorical** (*nominal or ordinal*)
 - Example: **Level of education** (e.g. high school, college, etc.) and **SES** (high, middle, and low)
- 2** Independent **categorical** (*nominal or ordinal*) & Dependent **continuous** (*interval-ratio*)
 - Example: **Level of education** (e.g. high school, college, etc.) vs **income** (e.g. annual dollar amount)
- 3** Independent **continuous** (*interval-ratio*) & **Dependent continuous** (*interval-ratio*)
 - Example: **Experience** (e.g. number of years on the job) & **income** (e.g. annual dollar amount)

Types of Bivariate Analysis

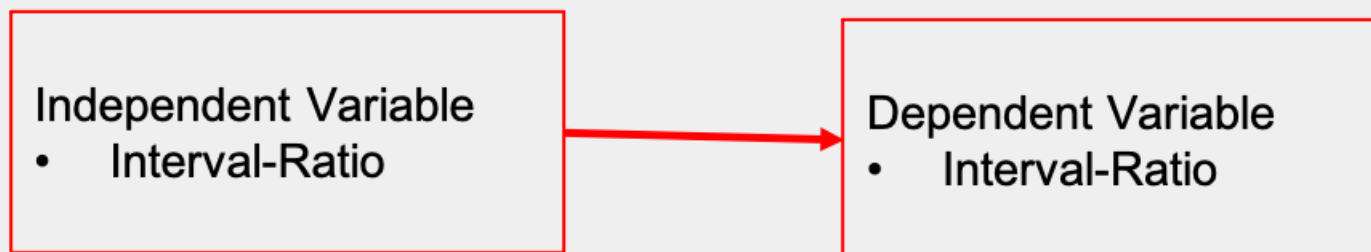


1 Contingency tables (Crosstabs): Relationship between two categorical variables



2 Comparing means: Relationship between a categorical independent and a continuous dependent variable

- Compare means of a continuous variable across different groups of the categorical independent variable



3 Correlation & simple regression: Relationship between two interval-ratio level variables

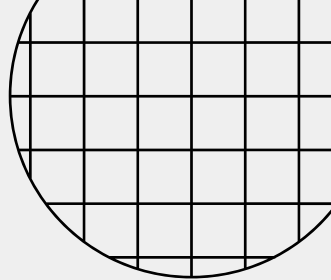
We display the values of two different variables at the same time in table (matrix) format

Known as:

- **Contingency tables**
- **Cross-tabulations/Crosstabs**

Bivariate Analysis of Categorical Variables

Example 1: College Major and Job



case_id	job	college_major
1	Social Worker	Other
2	Not Social Worker	Other
3	Not Social Worker	Social Work
4	Social Worker	Social Work
5	Not Social Worker	Other
6	Not Social Worker	Social Work
7	Not Social Worker	Other
8	Not Social Worker	Other
9	Social Worker	Social Work
10	Social Worker	Social Work
11	Not Social Worker	Other
12	Social Worker	Other
13	Social Worker	Social Work
14	Not Social Worker	Other
15	Social Worker	Social Work

Here is a snippet of the data

Here is a frequency table of the data for both variables

Job	Frequency
Social Worker	40
Not Social Worker	60
Total	100

College Major	Frequency
Social Work	55
Other	45
Total	100

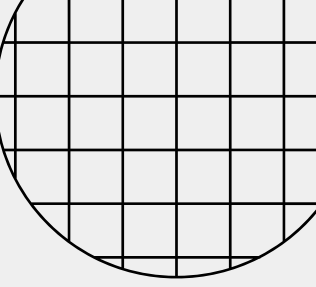
	College Major		
Job	Social Work	Other	Total
	30	10	40
	25	35	60
Total	55	45	100

Row Marginals

Column Marginals

Here is a contingency table of the data for both variables

Example 1: College Major and Job



	College Major		
Job	Social Work	Other	Total
Social Worker	30 (55%)	10 (22%)	40 (40%)
Not Social Worker	25 (45%)	35 (78%)	60 (60%)
Total	55 (100%)	45 (100%)	100 (100%)

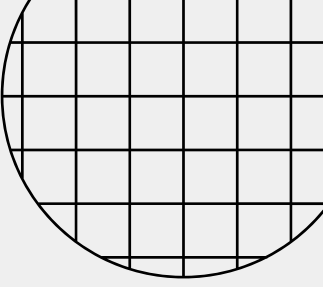
Here is the same contingency table of the data for both variables, but using **percentages** to standardize comparisons

We standardized across the independent variable **college major** to compare people in different majors

Social work students are more likely to be employed as a social worker

- 55% of social work students work students in sample are working as social workers (compared to only 22% of other-major students working as a social worker).

Example 1: College Major and Job



	College Major		
Job	Social Work	Other	Total
Social Worker	30 (75%)	10 (25%)	40 (100%)
Not Social Worker	25 (42%)	35 (58%)	60 (100%)
Total	55 (55%)	45 (45%)	100 (100%)

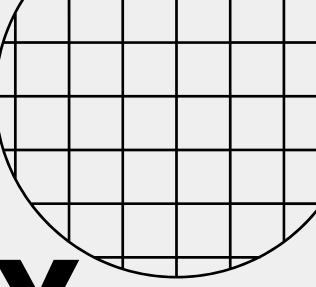
Here is the same contingency table of the data for both variables, but using **percentages** to standardize comparisons

We standardized across the dependent variable **job** to compare people in different careers

Social workers tend to most likely have social work degrees

- 75% of social workers got their degrees in social work (compared to only 25% of social workers getting degrees in other majors).

Example 2: Political Ideology and the Death Penalty



Opinion on death penalty	Political ideology			Total
	Liberal	Moderate	Conservative	
Favor	55.4%	70.1%	75.1%	462 (68.1%)
Oppose	44.6%	29.9%	24.9%	216 (31.9%)
Total	100%	100%	100%	678 (100%)

We standardized across the independent variable **political ideology** to compare people of different political stances

Liberal individuals are overrepresented in opposition to the death penalty

- 44.6% of liberals oppose death penalty (compared to 29.9% of moderates and 24.9% of conservatives who oppose death penalty).